OOPS Assignment-4

Name- Puneet Singh

Roll No- 1024150037

Q1.

main.cpp

```cpp
// Puneet Singh, 1024150037

#include <iostream>
using namespace std;

class Rectangle {
    int length, breadth;

public:
    // (a) Default constructor
    Rectangle() {
        length = 0;
        breadth = 0;
    }

    // (b) Two parameter constructor
    Rectangle(int l, int b) {
        length = l;
        breadth = b;
    }

    // (c) One parameter constructor
    Rectangle(int x) {
        length = x;
        breadth = x;
    }

    int area() {
        return length * breadth;
    }
};

int main() {
        cout << "Puneet Singh, 1024150037" << endl;

    Rectangle r1;         // no parameter
    Rectangle r2(5);      // one parameter
    Rectangle r3(4, 6);   // two parameters

    cout << "Area r1: " << r1.area() << endl;
    cout << "Area r2: " << r2.area() << endl;
    cout << "Area r3: " << r3.area() << endl;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Area r1: 0
Area r2: 25
Area r3: 24


=== Code Execution Successful ===
```

Q2.

main.cpp

```cpp
// Puneet Singh, 1024150037

#include <iostream>
using namespace std;

class Rectangle {
    int length, breadth;

public:
    // Constructor with default arguments
    Rectangle(int l = 0, int b = 0) {
        length = l;
        breadth = b;
    }

    int area() {
        return length * breadth;
    }

    // Destructor
    ~Rectangle() {
        cout << "Destructor called for Rectangle\n";
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    // Array of objects
    Rectangle r[3] = { Rectangle(), Rectangle(5), Rectangle(4, 6) };

    for(int i = 0; i < 3; i++) {
        cout << "Area of object " << i+1 << ": " << r[i].area() <<
            endl;
    }

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Area of object 1: 0
Area of object 2: 0
Area of object 3: 24
Destructor called for Rectangle
Destructor called for Rectangle
Destructor called for Rectangle


=== Code Execution Successful ===
```

Q3.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Demo {
public:
    // Constructor
    Demo() {
        cout << "Constructor called" << endl;
    }

    // Destructor (no parameter, no return type, only one allowed)
    ~Demo() {
        cout << "Destructor called (~Demo)" << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    Demo d1;  // Object created

    cout << "\nVerifying Destructor Properties:\n";
    cout << "1. Destructor name starts with ~ and matches class name (~Demo)\n";
    cout << "2. Only one destructor allowed per class\n";
    cout << "3. Destructor takes no parameters\n";
    cout << "4. Destructor has no return type\n";

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Constructor called

Verifying Destructor Properties:
1. Destructor name starts with ~ and matches class name (~Demo)
2. Only one destructor allowed per class
3. Destructor takes no parameters
4. Destructor has no return type
Destructor called (~Demo)


=== Code Execution Successful ===
```

Q4.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Demo {
public:
    int x;

    Demo(int val) {
        x = val;
        cout << "Object created: " << x << endl;
    }

    ~Demo() {
        cout << "Object destroyed: " << x << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    int *p = new int;
    *p = 10;
    cout << *p << endl;
    delete p;

    float *f = new float;
    *f = 5.5;
    cout << *f << endl;
    delete f;

    int *arr = new int[3];
    arr[0] = 1; arr[1] = 2; arr[2] = 3;
    for(int i = 0; i < 3; i++)
        cout << arr[i] << " ";
    cout << endl;
    delete[] arr;

    float *farr = new float[3];
    farr[0] = 1.1; farr[1] = 2.2; farr[2] = 3.3;
    for(int i = 0; i < 3; i++)
        cout << farr[i] << " ";
    cout << endl;
    delete[] farr;

    Demo *d = new Demo(100);
    delete d;

    Demo *darr = new Demo[2]{ Demo(1), Demo(2) };
    delete[] darr;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
10
5.5
1 2 3
1.1 2.2 3.3
Object created: 100
Object destroyed: 100
Object created: 1
Object created: 2
Object destroyed: 2
Object destroyed: 1


=== Code Execution Successful ===
```

Q5.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Demo {
public:
    int x;

    Demo(int a) {
        x = a;
    }

    Demo(const Demo &d) {
        x = d.x;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    Demo d1(10);
    Demo d2 = d1;

    cout << d1.x << endl;
    cout << d2.x << endl;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
10
10


=== Code Execution Successful ===
```

Q6.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Test {
    int a;

public:
    Test(int x) {
        a = x;
    }

    friend void show(Test t);
};

void show(Test t) {
    cout << t.a << endl;
}

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    Test t1(50);
    show(t1);

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
50


=== Code Execution Successful ===
```

Q7.

main.cpp

```cpp
#include <iostream>
using namespace std;

class A {
public:
    int x;
};

class B : public A {
public:
    int y;

    void display() {
        cout << x + y << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    B obj;
    obj.x = 10;
    obj.y = 20;
    obj.display();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
30


=== Code Execution Successful ===
```

OOPS Assignment- 5

Name- Puneet Singh

Roll No- 1024150037

Q1.

main.cpp

```cpp
#include <iostream>
using namespace std;

class A {
public:
    void showA() {
        cout << "Base class" << endl;
    }
};

class B : public A {
public:
    void showB() {
        cout << "Derived class" << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    A a;
    B b;

    a.showA();
    b.showA();
    b.showB();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Base class
Base class
Derived class


=== Code Execution Successful ===
```

Q2.

main.cpp

```cpp
#include <iostream>
using namespace std;

class A {
protected:
    int x;
};

class B : public A {
public:
    void set() {
        x = 10;
    }
    void show() {
        cout << x << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    B obj;
    obj.set();
    obj.show();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
10


=== Code Execution Successful ===
```

Q3.

main.cpp

```cpp
#include <iostream>
using namespace std;

class A {
public:
    int x;
protected:
    int y;
private:
    int z;
};

class B : public A {
public:
    void show() {
        x = 1;
        y = 2;
        cout << x + y << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    B obj;
    obj.show();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
3


=== Code Execution Successful ===
```

Q4.

main.cpp

```cpp
#include <iostream>
using namespace std;

class A {
public:
    A() { cout << "Base Constructor\n"; }
    ~A() { cout << "Base Destructor\n"; }
};

class B : public A {
public:
    B() { cout << "Derived Constructor\n"; }
    ~B() { cout << "Derived Destructor\n"; }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    B obj;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Base Constructor
Derived Constructor
Derived Destructor
Base Destructor


=== Code Execution Successful ===
```

Q5.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Book {
public:
    string title, author;
    int price;

    void set(string t, string a, int p) {
        title = t; author = a; price = p;
    }
};

class Textbook : public Book {
public:
    string subject;

    void setSub(string s) {
        subject = s;
    }

    void show() {
        cout << title << " " << author << " " << price << " " << subject << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    Textbook t;
    t.set("Math", "ABC", 500);
    t.setSub("Algebra");
    t.show();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Math ABC 500 Algebra


=== Code Execution Successful ===
```

Q6.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Speedometer {
public:
    int speed;
};

class FuelGauge {
public:
    int fuel;
};

class Thermometer {
public:
    int temp;
};

class CarDashboard : public Speedometer, public FuelGauge, public Thermometer {
public:
    void show() {
        cout << speed << " " << fuel << " " << temp << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    CarDashboard c;
    c.speed = 80;
    c.fuel = 50;
    c.temp = 30;

    c.show();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
80 50 30


=== Code Execution Successful ===
```

Q7.

main.cpp

```cpp
#include <iostream>
using namespace std;

class LibraryUser {
public:
    string name;
    int id;
};

class Student : public LibraryUser {
public:
    int grade;
};

class Teacher : public LibraryUser {
public:
    string dept;
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    Student s;
    Teacher t;

    s.name = "A"; s.id = 1; s.grade = 10;
    t.name = "B"; t.id = 2; t.dept = "CS";

    cout << s.name << " " << s.grade << endl;
    cout << t.name << " " << t.dept << endl;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
A 10
B CS


=== Code Execution Successful ===
```

Q8.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Vehicle {
public:
    string make;
};

class Truck : public Vehicle {
public:
    int load;
};

class RefrigeratedTruck : public Truck {
public:
    int temp;
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    RefrigeratedTruck r;
    r.make = "Tata";
    r.load = 1000;
    r.temp = -5;

    cout << r.make << " " << r.load << " " << r.temp << endl;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Tata 1000 -5


=== Code Execution Successful ===
```

OOPS Assignment- 6

Name- Puneet Singh

Roll No- 1024150037

Q1.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Book {
public:
    string title, author, isbn;
};

class Library {
public:
    Book b[10];
    int count = 0;

    bool addNewBook(string &title, string &author, string &isbn) {
        if(count >= 10) return false;
        b[count].title = title;
        b[count].author = author;
        b[count].isbn = isbn;
        count++;
        return true;
    }

    bool removeBooks(string &isbn) {
        for(int i=0;i<count;i++) {
            if(b[i].isbn == isbn) {
                for(int j=i;j<count-1;j++)
                    b[j] = b[j+1];
                count--;
                return true;
            }
        }
        return false;
    }

    void displayDetails() {
        for(int i=0;i<count;i++)
            cout << b[i].title << " " << b[i].author << " " << b[i].isbn << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    Library l;
    string t,a,i;

    t="A"; a="AA"; i="1"; l.addNewBook(t,a,i);
    t="B"; a="BB"; i="2"; l.addNewBook(t,a,i);
    t="C"; a="CC"; i="3"; l.addNewBook(t,a,i);
    t="D"; a="DD"; i="4"; l.addNewBook(t,a,i);
    t="E"; a="EE"; i="5"; l.addNewBook(t,a,i);

    l.removeBooks(i);
    l.displayDetails();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
A AA 1
B BB 2
C CC 3
D DD 4


=== Code Execution Successful ===
```

Q2.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Book {
public:
    string title, author, isbn;

    Book() {}

    Book(string title, string author, string isbn) {
        this->title = title;
        this->author = author;
        this->isbn = isbn;
    }

    Book(const Book &b) {
        this->title = b.title;
        this->author = b.author;
        this->isbn = b.isbn;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    Book b1("A","AA","1");
    Book b2 = b1;

    cout << b2.title << endl;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
A


=== Code Execution Successful ===
```

Q3.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Account {
public:
    const long acc_no;
    long tid;
    string type;
    double balance;

    Account(long a, double b) : acc_no(a) {
        balance = b;
        tid = 0;
    }

    long depositAmount(const long to, const long from, const double amt) {
        balance += amt;
        type = "credit";
        return ++tid;
    }

    long creditAmount(const long to, const long from, const double amt) {
        balance -= amt;
        type = "debit";
        return ++tid;
    }

    void displayDetails() const {
        cout << acc_no << " " << balance << " " << type << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    Account a1(101,1000);
    a1.depositAmount(101,0,500);
    a1.creditAmount(0,101,200);
    a1.displayDetails();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
101 1300 debit


=== Code Execution Successful ===
```

Q4.

main.cpp

```cpp
#include <iostream>
using namespace std;

class B;

class A {
public:
    int x;
    A(int a){ x=a; }
    friend int add(A,B);
};

class B {
public:
    int y;
    B(int b){ y=b; }
    friend int add(A,B);
};

int add(A a, B b) {
    return a.x + b.y;
}

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    A a(10);
    B b(20);

    cout << add(a,b) << endl;

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
30


=== Code Execution Successful ===
```

Q5.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Complex {
public:
    int r,i;

    Complex(int a=0,int b=0){ r=a; i=b; }

    Complex(const Complex &c){ r=c.r; i=c.i; }

    void display(){
        cout << r << "+" << i << "i" << endl;
    }

    friend Complex sum(Complex,Complex);
};

Complex sum(Complex c1, Complex c2){
    return Complex(c1.r+c2.r, c1.i+c2.i);
}

int main() {
    cout << "Puneet Singh, 1024150037" << endl;

    Complex c1(2,3), c2(4,5);
    Complex c3 = sum(c1,c2);

    c3.display();

    return 0;
}
```

Output

```
Puneet Singh, 1024150037
6+8i


=== Code Execution Successful ===
```

OOPS Assignment- 7

Name- Puneet Singh

Roll No- 1024150037

Q1.

main.cpp

```cpp
// 1
#include <iostream>
using namespace std;

class polygon {
protected:
    int width, height;
public:
    void set_value(int w, int h) {
        width = w;
        height = h;
    }
};

class Rectangle : public polygon {
public:
    int calculate_area() {
        return width * height;
    }
};

class Triangle : public polygon {
public:
    int calculate_area() {
        return (width * height) / 2;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    Rectangle rect;
    Triangle tri;
    polygon *ptr1 = &rect;
    polygon *ptr2 = &tri;
    rect.set_value(10, 5);
    tri.set_value(10, 5);
    cout << "Rectangle Area: " << ((Rectangle*)ptr1)->calculate_area() << endl;
    cout << "Triangle Area: " << ((Triangle*)ptr2)->calculate_area() << endl;
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Rectangle Area: 50
Triangle Area: 25


=== Code Execution Successful ===
```

Q2.

main.cpp

```cpp
#include <iostream>
using namespace std;

class shape {
public:
    virtual void area() = 0;
    virtual void display() = 0;
};

class circle : public shape {
    float radius;
public:
    circle(float r) : radius(r) {}
    void area() {
        cout << "Circle Area: " << 3.14159 * radius * radius << endl;
    }
    void display() {
        cout << "Shape: Circle" << endl;
    }
};

class rectangle : public shape {
    float length, breadth;
public:
    rectangle(float l, float b) : length(l), breadth(b) {}
    void area() {
        cout << "Rectangle Area: " << length * breadth << endl;
    }
    void display() {
        cout << "Shape: Rectangle" << endl;
    }
};

class triangle : public shape {
    float base, height;
public:
    triangle(float b, float h) : base(b), height(h) {}
    void area() {
        cout << "Triangle Area: " << 0.5 * base * height << endl;
    }
    void display() {
        cout << "Shape: Triangle" << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    circle c(3);
    rectangle r(4, 6);
    triangle t(3, 8);
    c.display();
    c.area();
    r.display();
    r.area();
    t.display();
    t.area();
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Shape: Circle
Circle Area: 78.5397
Shape: Rectangle
Rectangle Area: 24
Shape: Triangle
Triangle Area: 12


--- Code Execution Successful ---
```

Q3.

main.cpp

```cpp
// 3
#include <iostream>
#include <cmath>
using namespace std;

float area(float base, float height) {
    return 0.5 * base * height;
}

float area(float side) {
    return (sqrt(3) / 4) * side * side;
}

float area(float side1, float side2, float side3) {
    float s = (side1 + side2 + side3) / 2;
    return sqrt(s * (s - side1) * (s - side2) * (s - side3));
}

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    cout << "Right Angle Triangle: " << area(5, 8) << endl;
    cout << "Equilateral Triangle: " << area(6) << endl;
    cout << "Isosceles Triangle: " << area(5, 5, 6) << endl;
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Shape: Circle
Circle Area: 78.5397
Shape: Rectangle
Rectangle Area: 24
Shape: Triangle
Triangle Area: 12


=== Code Execution Successful ===
```

Q4.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Student {
public:
    virtual void display() = 0;
};

class Engineering : public Student {
public:
    void display() {
        cout << "Engineering Student" << endl;
    }
};

class Medicine : public Student {
public:
    void display() {
        cout << "Medicine Student" << endl;
    }
};

class Science : public Student {
public:
    void display() {
        cout << "Science Student" << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    Student *arr[3];
    Engineering e;
    Medicine m;
    Science s;
    arr[0] = &e;
    arr[1] = &m;
    arr[2] = &s;
    for(int i = 0; i < 3; i++) {
        arr[i]->display();
    }
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Engineering Student
Medicine Student
Science Student


=== Code Execution Successful ===
```

Q5.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Time {
    int h, m, s;
public:
    Time() : h(0), m(0), s(0) {}
    Time(int hours, int minutes, int seconds) : h(hours), m(minutes), s(seconds) {}
    Time operator+(const Time &t) {
        Time temp;
        temp.s = s + t.s;
        temp.m = m + t.m + temp.s / 60;
        temp.s %= 60;
        temp.h = h + t.h + temp.m / 60;
        temp.m %= 60;
        return temp;
    }
    void show() {
        cout << h << ":" << m << ":" << s << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    Time t1(5,15,34), t2(9,53,58), t3;
    t3 = t1 + t2;
    t3.show();
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
15:9:32


=== Code Execution Successful ===
```

Q6.

main.cpp

```cpp
#include <iostream>
#include <cstring>
using namespace std;

class STRING {
    char str[100];
public:
    STRING() {
        str[0] = '\0';
    }
    STRING(const char *s) {
        strcpy(str, s);
    }
    bool operator==(const STRING &s) {
        return strcmp(str, s.str) == 0;
    }
    STRING operator+(const STRING &s) {
        STRING temp;
        strcpy(temp.str, str);
        strcat(temp.str, s.str);
        return temp;
    }
    void display() {
        cout << str << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    STRING s1("Hello"), s2("Hello"), s3("World");
    if(s1 == s2) {
        cout << "Strings are equal" << endl;
    }
    STRING s4 = s1 + s3;
    s4.display();
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Strings are equal
HelloWorld


=== Code Execution Successful ===
```

Q7.

main.cpp

```cpp
#include <iostream>
using namespace std;

class matrix {
    int arr[10][10];
    int r, c;
public:
    matrix(int rows = 0, int cols = 0) : r(rows), c(cols) {
        for(int i = 0; i < r; i++)
            for(int j = 0; j < c; j++)
                arr[i][j] = 0;
    }
    void input() {
        for(int i = 0; i < r; i++)
            for(int j = 0; j < c; j++)
                cin >> arr[i][j];
    }
    friend matrix operator*(const matrix &a, const matrix &b);
    void display() {
        for(int i = 0; i < r; i++) {
            for(int j = 0; j < c; j++)
                cout << arr[i][j] << " ";
            cout << endl;
        }
    }
};

matrix operator*(const matrix &a, const matrix &b) {
    matrix result(a.r, b.c);
    for(int i = 0; i < a.r; i++)
        for(int j = 0; j < b.c; j++)
            for(int k = 0; k < a.c; k++)
                result.arr[i][j] += a.arr[i][k] * b.arr[k][j];
    return result;
}

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    matrix m1(2,2), m2(2,2), m3;
    cout << "Enter first 2x2 matrix:" << endl;
    m1.input();
    cout << "Enter second 2x2 matrix:" << endl;
    m2.input();
    m3 = m1 * m2;
    cout << "Product matrix:" << endl;
    m3.display();
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Enter first 2x2 matrix:
5
4
5
3
Enter second 2x2 matrix:
4
1
7
8
Product matrix:
48 37
41 29


=== Code Execution Successful ===
```

Q8.

main.cpp

```cpp
// 8
#include <iostream>
#include <cstdlib>
using namespace std;

class Array {
    int arr[10];
public:
    int &operator[](int index) {
        if(index < 0 || index >= 10) {
            cout << "Index out of bounds!" << endl;
            exit(1);
        }
        return arr[index];
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    Array a;
    a[0] = 5;
    cout << a[0] << endl;
    a[15] = 10;
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
5
Index out of bounds!


=== Code Exited With Errors ===
```

Q9.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Functor {
public:
    void operator()(int a, int b, int c) {
        cout << "Sum: " << a + b + c << endl;
    }
    void operator()(int a, int b, int c, int d) {
        cout << "Sum: " << a + b + c + d << endl;
    }
    void operator()(int a, int b) {
        cout << "Sum: " << a + b << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    Functor f;
    f(1, 2);
    f(1, 2, 3);
    f(1, 2, 3, 4);
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Sum: 3
Sum: 6
Sum: 10


=== Code Execution Successful ===
```

Q10.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Point {
    int x, y;
public:
    friend istream &operator>>(istream &in, Point &p);
    friend ostream &operator<<(ostream &out, const Point &p);
};

istream &operator>>(istream &in, Point &p) {
    in >> p.x >> p.y;
    return in;
}

ostream &operator<<(ostream &out, const Point &p) {
    out << "(" << p.x << ", " << p.y << ")";
    return out;
}

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    Point p;
    cout << "Enter x and y: ";
    cin >> p;
    cout << "You entered: " << p << endl;
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Enter x and y: 5
3
You entered: (5, 3)


=== Code Execution Successful ===
```

Q11.

main.cpp

```cpp
#include <iostream>
using namespace std;

class Test {
    float value;
public:
    Test(float f) {
        value = f;
    }
    void display() {
        cout << "Value: " << value << endl;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    float f = 3.14;
    Test t = f;
    t.display();
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Value: 3.14


=== Code Execution Successful ===
```

Q12.

main.cpp

```cpp
// 12
#include <iostream>
using namespace std;

class Test {
    float value;
public:
    Test(float v = 0) : value(v) {}
    operator float() {
        return value;
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    Test t(3.14);
    float f = t;
    cout << f << endl;
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
3.14


=== Code Execution Successful ===
```

Q13.

main.cpp

```cpp
// 13
#include <iostream>
#include <cmath>
using namespace std;

class Cartesian {
    float x, y;
public:
    Cartesian(float a = 0, float b = 0) : x(a), y(b) {}
    void show() {
        cout << "Cartesian: (" << x << ", " << y << ")" << endl;
    }
};

class Polar {
    float radius, angle;
public:
    Polar(float r, float a) : radius(r), angle(a) {}
    operator Cartesian() {
        float x = radius * cos(angle * 3.14159 / 180);
        float y = radius * sin(angle * 3.14159 / 180);
        return Cartesian(x, y);
    }
};

int main() {
    cout << "Puneet Singh, 1024150037" << endl;
    Polar p(10, 5);
    Cartesian c = p;
    c.show();
    return 0;
}
```

Output

```
Puneet Singh, 1024150037
Cartesian: (9.96195, 0.871557)


=== Code Execution Successful ===
```